

# Ricardo Yarza

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## APPOINTMENTS

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Chancellor’s Postdoctoral Fellow, University of California, Los Angeles 2026–

## EDUCATION

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PhD Astronomy and Astrophysics, University of California, Santa Cruz 2026  
Advisor: Professor Enrico Ramirez-Ruiz

BS Engineering Physics, Tecnológico de Monterrey 2019

Study abroad, University of Illinois at Urbana–Champaign 2017–2018  
Advisor: Professor Charles Gammie

## PUBLICATIONS

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Online lists: [NASA ADS](#), [Google Scholar](#), [ORCID](#).

Underline indicates mentored student.

In press or under review journal articles:

2. Twum, A. A. G., Vigna-Gómez, A., MacLeod, M., Everson, R. W., et al. incl. **Yarza, R.** 2026, [arXiv:2602.10211](#)  
*The Landscape of Unstable Mass Transfer in Interacting Binaries and Its Imprint on the Population of Luminous Red Novae*
1. Soares-Furtado, M., Limbach, M. A., Vanderburg, A., Bally, J., et al. incl. **Yarza, R.** 2024, [arXiv:2406.01492](#)  
*The TEMPO Survey II: Science Cases Leveraged from a Proposed 30-Day Time Domain Survey of the Orion Nebula with the Nancy Grace Roman Space Telescope*

Published journal articles:

17. Jenkins, S., Vanderburg, A., Mang, J., Becker, J., et al. incl. **Yarza, R.** 2026, [ApJ](#), 1006, 241  
*Atmospheric Signatures of Common-envelope Evolution in White Dwarf Planets*
16. Kotten, B., Soares-Furtado, M., **Yarza, R.**, Nine, A. C., et al. 2026, [ApJ](#), 1004, 193  
*Lithium Enrichment in a Subgiant Star with a Brown Dwarf Companion: A Planetary Engulfment Candidate*
15. Lane, K. T., Stephan, A. P., Soares-Furtado, M., Stassun, K. G., & **Yarza, R.** 2026, [ApJ](#), 1003, 67  
*Observable Metal Pollution in Main-sequence Stars: Simulations of Rocky Planets Engulfed by Stars in the 0.5 to 1.4 M<sub>⊙</sub> Range*
14. **Yarza, R.**, MacLeod, M., Idini, B., Murray-Clay, R., & Ramirez-Ruiz, E. 2025, [ApJL](#), 995, L48  
*Evolution of the ZTF SLRN-2020 Star-Planet Merger*

13. Bush, R. C., Wu, S. C., Everson, R. W., **Yarza, R.**, et al. 2025, [ApJL, 990, L7](#)  
*Black Hole Survival Guide: Searching for Stars in the Galactic Center that Endure Partial Tidal Disruption*
12. Liu, C., **Yarza, R.**, & Ramirez-Ruiz, E. 2025, [ApJ, 979, 40](#)  
*Repeating Partial Tidal Encounters of Sun-like Stars Leading to Their Complete Disruption*
11. Farias, D., Gall, C., Narayan, G., Rest, S., et al. incl. **Yarza, R.** 2024, [ApJ, 977, 152](#)  
*SN 2021foa: The “Flip-flop” Type II<sub>n</sub>/Ib<sub>n</sub> Supernova*
10. Rosselli-Calderon, A., **Yarza, R.**, Murguia-Berthier, A., Rohoza, V., et al. 2024, [ApJ, 977, 16](#)  
*The Evolution of Binaries Embedded Within Common Envelopes*
9. Hutchinson-Smith, T., Everson, R. W., Twum, A. A., Batta, A., et al. incl. **Yarza, R.** 2024, [ApJ, 977, 196](#)  
*Rethinking Thorne–Żytkow Object Formation: The Fate of X-Ray Binary LMC X-4 and Implications for Ultra-long Gamma-Ray Bursts*
8. Ransome, C. L., Villar, V. A., Tartaglia, A., Gonzalez, S. J., et al. incl. **Yarza, R.** 2024, [ApJ, 965, 93](#)  
*SN 2023ixf in Messier 101: The Twilight Years of the Progenitor as Seen by Pan-STARRS*
7. Capistrant, B. K., Soares-Furtado, M., Vanderburg, A., Jankowski, A., et al. incl. **Yarza, R.** 2024, [AJ, 167, 54](#)  
*TESS Hunt for Young and Maturing Exoplanets (THYME). XI. An Earth-sized Planet Orbiting a Nearby, Solar-like Host in the 400 Myr Ursa Major Moving Group*
6. **Yarza, R.**, Razo-López, N. B., Murguia-Berthier, A., Everson, R. W., et al. 2023, [ApJ, 954, 176](#)  
*Hydrodynamics and Survivability during Post-main-sequence Planetary Engulfment*
5. Aleo, P. D., Malanchev, K., Sharief, S., Jones, D. O., et al. incl. **Yarza, R.** 2023, [ApJS, 266, 9](#)  
*The Young Supernova Experiment Data Release 1 (YSE DR1): Light Curves and Photometric Classification of 1975 Supernovae*
4. Liu, C., Mockler, B., Ramirez-Ruiz, E., **Yarza, R.**, et al. 2023, [ApJ, 944, 184](#)  
*Tidal Disruption Events from Eccentric Orbits and Lessons Learned from the Noteworthy ASASSN-14ko*
3. Wong, G. N., Prather, B. S., Dhruv, V., Ryan, B. R., et al. incl. **Yarza, R.** 2022, [ApJS, 259, 64](#)  
*PATOKA: Simulating Electromagnetic Observables of Black Hole Accretion*
2. **Yarza, R.**, Wong, G. N., Ryan, B. R., & Gammie, C. F. 2020, [ApJ, 898, 50](#)  
*Bremsstrahlung in GRMHD Models of Accreting Black Holes*
1. Tovmassian, G., Szkody, P., **Yarza, R.**, & Kennedy, M. 2018, [ApJ, 863, 47](#)  
*K2 Study of the Magnetic Precataclysmic Variable V1082 Sagittarius*

## AWARDS

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|                                                                                                                 |           |
|-----------------------------------------------------------------------------------------------------------------|-----------|
| Chancellor's Postdoctoral Fellowship<br>University of California, Los Angeles                                   | 2026–2028 |
| Chancellor's Dissertation-Year Fellowship<br>UC Santa Cruz                                                      | Fall 2025 |
| Future Investigators in NASA Earth and Space Science and Technology<br>NASA                                     | 2022–2025 |
| Doctoral Fellowship<br>University of California Institute for Mexico and the United States                      | 2020–2025 |
| Raynor L. Duncombe Student Research Prize<br>Division on Dynamical Astronomy, American Astronomical Society     | 2023      |
| Frontera Computational Science Fellowship<br>Texas Advanced Computing Center, The University of Texas at Austin | 2022–2023 |
| Regents Fellowship<br>UC Santa Cruz                                                                             | 2019      |

## TALKS

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|------------------------------------------------------------------------|------------|
| Theoretical Astrophysics Center Seminar, UC Berkeley                   | Dec 2025   |
| 48th Bay Area Exoplanet Meeting / 7th OWL Exoplanet Summer Program     | Jul 2025   |
| ITC Luncheon Talk, Center for Astrophysics, Harvard & Smithsonian      | May 2024   |
| Meeting of the American Physical Society                               | April 2024 |
| 54th meeting of the AAS Division on Dynamical Astronomy                | May 2023   |
| Texas Advanced Computing Center, The University of Texas at Austin     | May 2023   |
| Exoplanet Systems and Stellar Life Cycles, Aspen Center of Physics     | Mar 2023   |
| Friday Lunch Time Astrophysics Seminar (FLASH), UC Santa Cruz          | Dec 2022   |
| Seminar in Geophysical and Astrophysical Fluid Dynamics, UC Santa Cruz | Nov 2022   |
| Exoplanet Summer Program, Other Worlds Laboratory                      | July 2022  |
| Texas Advanced Computing Center, The University of Texas at Austin     | July 2022  |
| 240th meeting of the American Astronomical Society                     | June 2022  |

## SERVICE

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| Referee for <i>Astronomy &amp; Astrophysics</i> , <i>The Astrophysical Journal</i> , <i>Science Bulletin</i> |            |
| Session Chair, <i>Neutron Stars and Stellar Objects</i><br>Meeting of the American Physical Society          | April 2024 |
| Executive Secretary, NASA Astrophysics Theory Program review panel                                           | Oct 2023   |
| Designed and gave workshops on N-body simulation, data visualization, and <i>Gaia</i> EDR3                   |            |

## MENTORING

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Mentored students:

- Richard Truong, undergraduate at San Francisco State University
- Brooke Kotten, undergraduate at the University of Wisconsin, Madison  
Now graduate student at the University of Michigan
- Valeriia Rohoza, undergraduate at Northwestern  
Now graduate student at Columbia
- Alejandra Meza, high school student at Pacific Collegiate School  
Now physics undergraduate at UC Berkeley
- Alejandra Rosselli-Calderon, MS student at Brown  
Now graduate student at UC Santa Cruz

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|------------------------------------------------------------------------------------|--------------|
| Lamat program                                                                      | 2020–present |
| Society of Physics Students Mentoring Program                                      | 2021         |
| <a href="#">Young Scholars Program</a> : Derek Davis, junior at Urbana High School | Summer 2018  |

## TEACHING

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Teaching assistant, UC Santa Cruz:

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|-------------------------------------------------------------------------------------------------------|-------------|
| – Astronomy 118: <i>Physics of planetary systems</i><br>Taught one lecture; wrote and gave final exam | Spring 2022 |
| – Astronomy 116: <i>Physics of stars</i>                                                              | Spring 2021 |
| – Astronomy 119: <i>Introduction to scientific computing</i>                                          | Fall 2020   |
| – Astronomy 16: <i>Astrobiology: life in the universe</i>                                             | Spring 2020 |
| – Astronomy 6: <i>The space-age solar system</i>                                                      | Winter 2020 |

## PRESS

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|------------------------------------------------------------------------------------------------------------------------|----------|
| <a href="#">Discovery Alert: Earth-sized Planet Has a “Lava Hemisphere”</a><br>Chelsea Gohd, NASA                      | Jan 2024 |
| <a href="#">The Juicy Secrets of Stars That Eat Their Planets</a><br>Becky Ferreira, The New York Times                | Aug 2022 |
| <a href="#">Astronomers Reveal New Details of How Stars Devour Planets</a><br>Nola Taylor Tillman, Scientific American | Aug 2022 |